Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Brazil/ Van Ryn Dairy Farm LLC

Physical address of dairy:

Number and Street	City	County	Zip Code
18338 E Mariposa RD	Stockton	San Joaquin	95215

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 09/01/2004

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X205-X020-XX03-XXXX

B. OPERATORS

Brazil, Anthony Bartholomew

Operator name: Brazil, Anthony Bartholomew

Telephone no.: (209) 886-1061 Landline, (209) 985-2410 Cellular

Mailing Address Number and Street	City	State	Zip Code
11535 S Henry RD	Escalon	CA	95320

Van Ryn, William John

Operator name: Van Ryn, William John

Telephone no.: (209) 824-2284 Landline, (209) 482-2295 Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 159	Ripon	CA	95366

This operator is responsible for paying permit fees.

C. OWNERS

Brazil, Anthony Bartholomew

Legal owner name: Brazil, Anthony Bartholomew

Telephone no.: (209) 886-1061 Landline, (209) 985-2410 Cellular

Mailing Address Number and Street	City	State	Zip Code
11535 S Henry RD	Escalon	CA	95320

Van Ryn, William John

Legal owner name:	Van Ryn, William John	Telephone no.:	(209) 824-2284	(209) 482-2295
			Landline	Cellular
P.O. Box 159	Ripon		CA	95366
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	75	85	120	0
Number under roof	550	75	0	0	0	75
Maximum number	550	75	75	85	120	75
Average number	550	75	75	85	120	75
Avg live weight (lbs)	1,300	1,350	1,000	600		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 17,281.16 *tons per reporting period*

Total nitrogen from manure: 221,607.15 *lbs per reporting period*

After ammonia losses (30% loss applied): 155,125.01 *lbs per reporting period*

Total phosphorus from manure: 35,998.43 *lbs per reporting period*

Total potassium from manure: 104,607.37 *lbs per reporting period*

Total salt from manure: 276,213.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 2,100,000 *gallons*

Total nitrogen generated: 10,409.55 *lbs*

Total phosphorus generated: 1,682.35 *lbs*

Total potassium generated: 25,410.53 *lbs*

Total salt generated: 35,399.49 *lbs*

	2,100,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	2,100,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Dairy #1	Ground water
Domestic Dairy #2	Ground water
Irr. Ag Well #1	Ground water
Irr. Ag Well #2	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #1	10	10	1	both	X205-X020-XX03-XXXX
Field #2	10	10	1	both	X205-X020-XX03-XXXX
Field #3	15	15	1	both	X205-X020-XX03-XXXX
Field #4	15	15	1	both	X205-X020-XX03-XXXX
Field #5	90	90	6	both	X205-X020-XX03-XXXX
Field #6	50	50	4	both	X205-X020-XX03-XXXX
Totals for areas that were used for application	190	190	14		
Totals for areas that were not used for application					
Land application area totals	190	190	14		

B. CROPS AND HARVESTS

Field #1

Field name: Field #1

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	130.00 *ton*	Dry-weight		62.9	20,784.00	3,300.00	25,100.00		9.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.00	200.48	31.83	242.11	958.81

Field #2

Field name: Field #2

Field #2

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	140.00 *ton*	Dry-weight		75.0	19,904.00	2,700.00	30,500.00		10.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.00	139.33	18.90	213.50	721.70

Field #3

Field name: Field #3

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	202.50 *ton*	Dry-weight		75.0	19,904.00	2,700.00	30,500.00		10.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.50	134.35	18.23	205.88	695.93

Field #4

Field name: Field #4

Field #4

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	195.00 *ton*	Dry-weight		62.9	20,784.00	3,300.00	25,100.00		9.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.00	200.48	31.83	242.11	958.81

Field #5

Field name: Field #5

10/29/2022: Winter Forage

Crop: Winter Forage Acres planted: 90 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	520.00 *ton*	Dry-weight		69.6	16,800.00	3,900.00	49,900.00		13.20
05/10/2023	520.00 *ton*	Dry-weight		70.2	15,700.00	4,000.00	52,200.00		13.50
05/10/2023	130.00 *ton*	Dry-weight		75.6	18,700.00	4,500.00	61,700.00		14.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	142.80	33.60	210.00	0.00
Total actual harvest content	13.00	126.26	30.65	398.54	1,032.90

Field #5

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 90 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	1,140.00 *ton*	Dry-weight		68.0	13,700.00	2,500.00	16,800.00		8.20
10/17/2023	1,140.00 *ton*	Dry-weight		66.2	10,900.00	2,380.00	15,000.00		8.30
10/17/2023	285.00 *ton*	Dry-weight		69.6	13,500.00	2,480.00	15,600.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	246.40	44.80	296.80	0.00
Total actual harvest content	28.50	230.39	45.42	294.67	1,552.58

Field #6

Field name: Field #6

10/29/2022: Winter Forage

Crop: Winter Forage Acres planted: 50 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	390.00 *ton*	Dry-weight		75.6	18,700.00	4,500.00	61,700.00		14.80
05/10/2023	390.00 *ton*	Dry-weight		75.8	18,100.00	5,100.00	66,000.00		14.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	123.20	35.00	224.00	0.00
Total actual harvest content	15.60	139.51	36.38	484.02	1,114.53

Field #6

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	285.00 *ton*	Dry-weight		69.6	13,500.00	2,480.00	15,600.00		9.20
10/17/2023	1,140.00 *ton*	Dry-weight		68.2	12,100.00	2,190.00	14,400.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	257.60	0.00
Total actual harvest content	28.50	222.25	40.35	262.87	1,507.90

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #1 - 10/29/2022: Oats, silage-soft dough

Field name: Field #1

Crop: Oats, silage-soft dough — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	158.58	28.25	274.64	1,549.38	70.00 *ton*
Application event totals		158.58	28.25	274.64	1,549.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	89.22	14.42	217.80	303.42	180,000.00 *gal*
Application event totals		89.22	14.42	217.80	303.42	

Field #2 - 10/29/2022: Oats, silage-soft dough

Field name: Field #2

Crop: Oats, silage-soft dough — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	113.27	20.18	196.17	1,106.70	50.00 *ton*
Application event totals		113.27	20.18	196.17	1,106.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	59.48	9.61	145.20	202.28	120,000.00 *gal*
Application event totals		59.48	9.61	145.20	202.28	

Field #3 - 10/29/2022: Oats, silage-soft dough

Field name: Field #3

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	113.27	20.18	196.17	1,106.70	75.00 *ton*
Application event totals		113.27	20.18	196.17	1,106.70	

02/13/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	59.48	9.61	145.20	202.28	180,000.00 *gal*
Application event totals		59.48	9.61	145.20	202.28	

Field #4 - 10/29/2022: Oats, silage-soft dough

Field name: Field #4

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	158.58	28.25	274.64	1,549.38	105.00 *ton*
Application event totals		158.58	28.25	274.64	1,549.38	

02/14/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	59.48	9.61	145.20	202.28	180,000.00 *gal*
Application event totals		59.48	9.61	145.20	202.28	

Field #5 - 10/29/2022: Winter Forage

Field #5 - 10/29/2022: Winter Forage

Field name: Field #5

Crop: Winter Forage

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	113.27	20.18	196.17	1,106.70	450.00 *ton*
Application event totals		113.27	20.18	196.17	1,106.70	

02/14/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	49.57	8.01	121.00	168.57	900,000.00 *gal*
Application event totals		49.57	8.01	121.00	168.57	

Field #5 - 06/20/2023: Corn, silage

Field name: Field #5

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	205.38	51.31	72.64	0.00	1,170.00 *ton*
Application event totals		205.38	51.31	72.64	0.00	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.74	0.00	0.00	0.00	4,248,000.00 *gal*
Irr. Ag Well #2	Ground water	1.29	0.00	0.00	0.00	5,664,000.00 *gal*
Application event totals		2.03	0.00	0.00	0.00	

Field #5 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.72	0.00	0.00	0.00	4,104,000.00 *gal*
Irr. Ag Well #2	Ground water	1.25	0.00	0.00	0.00	5,472,000.00 *gal*
Application event totals		1.96	0.00	0.00	0.00	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	3,906,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	5,208,000.00 *gal*
Application event totals		1.87	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	3,906,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	5,208,000.00 *gal*
Application event totals		1.87	0.00	0.00	0.00	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	3,906,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	5,208,000.00 *gal*
Application event totals		1.87	0.00	0.00	0.00	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	3,906,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	5,208,000.00 *gal*
Application event totals		1.87	0.00	0.00	0.00	

Field #5 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	3,906,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	5,208,000.00 *gal*
Application event totals		1.87	0.00	0.00	0.00	

Field #6 - 10/29/2022: Winter Forage

Field name: Field #6

Crop: Winter Forage Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	113.27	20.18	196.17	1,106.70	250.00 *ton*
Application event totals		113.27	20.18	196.17	1,106.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	53.53	8.65	130.68	182.05	540,000.00 *gal*
Application event totals		53.53	8.65	130.68	182.05	

Field #6 - 06/20/2023: Corn, silage

Field name: Field #6

Crop: Corn, silage Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field #6 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Scraped material	205.38	51.31	72.64	0.00	650.00 *ton*
Application event totals		205.38	51.31	72.64	0.00	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.79	0.00	0.00	0.00	2,520,000.00 *gal*
Irr. Ag Well #2	Ground water	1.38	0.00	0.00	0.00	3,360,000.00 *gal*
Application event totals		2.17	0.00	0.00	0.00	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.73	0.00	0.00	0.00	2,340,000.00 *gal*
Irr. Ag Well #2	Ground water	1.28	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		2.02	0.00	0.00	0.00	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	2,178,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	2,904,000.00 *gal*
Application event totals		1.88	0.00	0.00	0.00	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	2,178,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	2,904,000.00 *gal*
Application event totals		1.88	0.00	0.00	0.00	

Field #6 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	2,178,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	2,904,000.00 *gal*
Application event totals		1.88	0.00	0.00	0.00	

09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	2,178,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	2,904,000.00 *gal*
Application event totals		1.88	0.00	0.00	0.00	

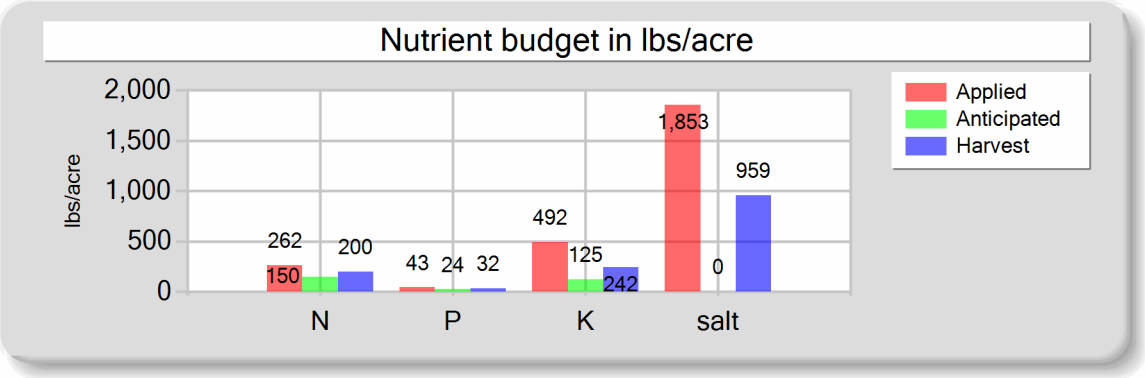
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irr. Ag Well #1	Ground water	0.68	0.00	0.00	0.00	2,178,000.00 *gal*
Irr. Ag Well #2	Ground water	1.19	0.00	0.00	0.00	2,904,000.00 *gal*
Application event totals		1.88	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field #1 - 10/29/2022: Oats, silage-soft dough

Field name: Field #1 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	158.58	28.25	274.64	1,549.38
Process wastewater	89.22	14.42	217.80	303.42
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	261.81	42.67	492.44	1,852.80
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	200.48	31.83	242.11	958.81
Nutrient balance	61.33	10.84	250.33	893.99
Applied to removed ratio	1.31	1.34	2.03	1.93

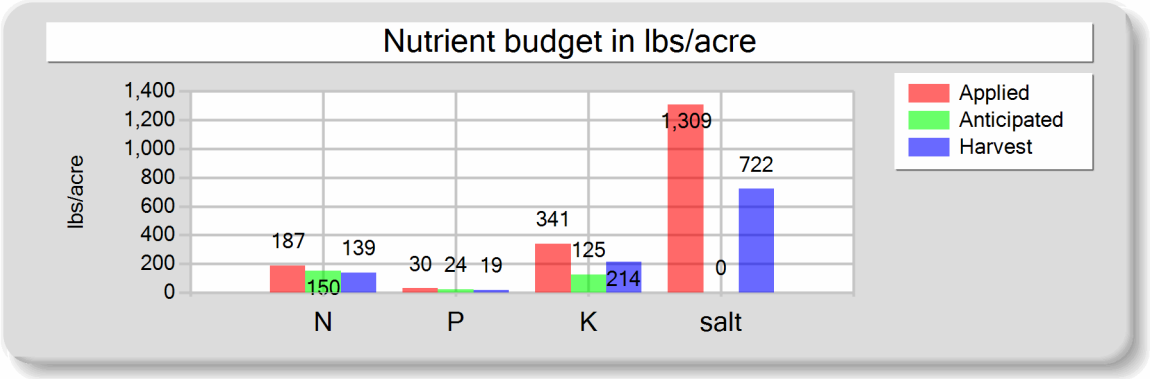
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
180,000.00 *gallons*
6.63 *acre-inches*
0.66 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #2 - 10/29/2022: Oats, silage-soft dough

Field name: Field #2 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	113.27	20.18	196.17	1,106.70
Process wastewater	59.48	9.61	145.20	202.28
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	186.76	29.79	341.37	1,308.98
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	139.33	18.90	213.50	721.70
Nutrient balance	47.43	10.89	127.87	587.28
Applied to removed ratio	1.34	1.58	1.60	1.81

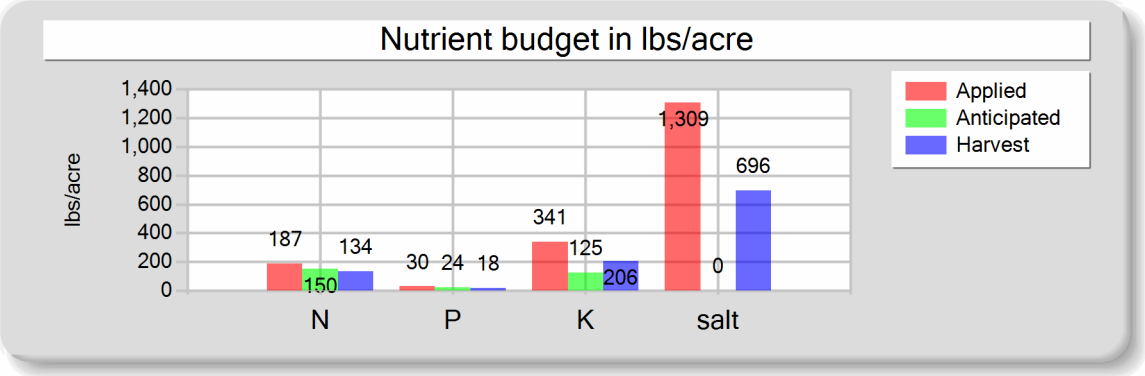
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
120,000.00 *gallons*
4.42 *acre-inches*
0.44 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #3 - 10/29/2022: Oats, silage-soft dough

Field name: Field #3 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	113.27	20.18	196.17	1,106.70
Process wastewater	59.48	9.61	145.20	202.28
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	186.76	29.79	341.37	1,308.98
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	134.35	18.23	205.88	695.93
Nutrient balance	52.41	11.57	135.50	613.06
Applied to removed ratio	1.39	1.63	1.66	1.88

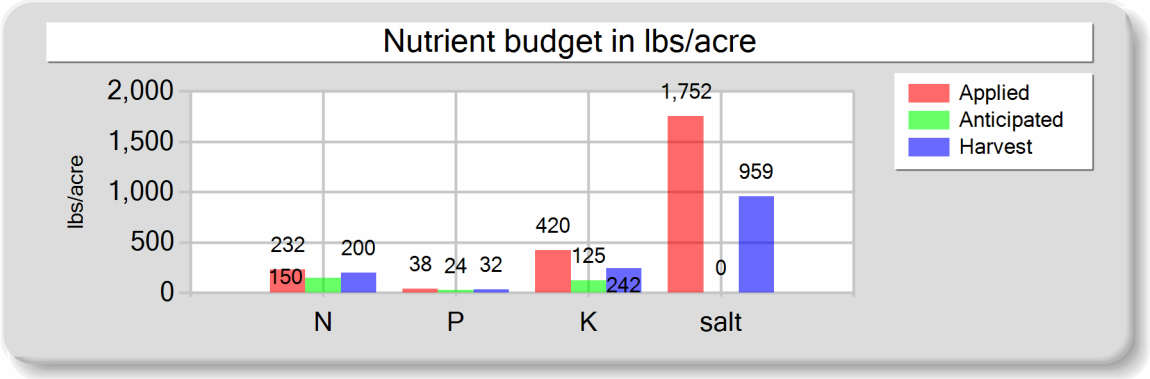
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
180,000.00 *gallons*
6.63 *acre-inches*
0.44 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #4 - 10/29/2022: Oats, silage-soft dough

Field name: Field #4 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	158.58	28.25	274.64	1,549.38
Process wastewater	59.48	9.61	145.20	202.28
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	232.07	37.87	419.84	1,751.66
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	200.48	31.83	242.11	958.81
Nutrient balance	31.58	6.04	177.72	792.85
Applied to removed ratio	1.16	1.19	1.73	1.83

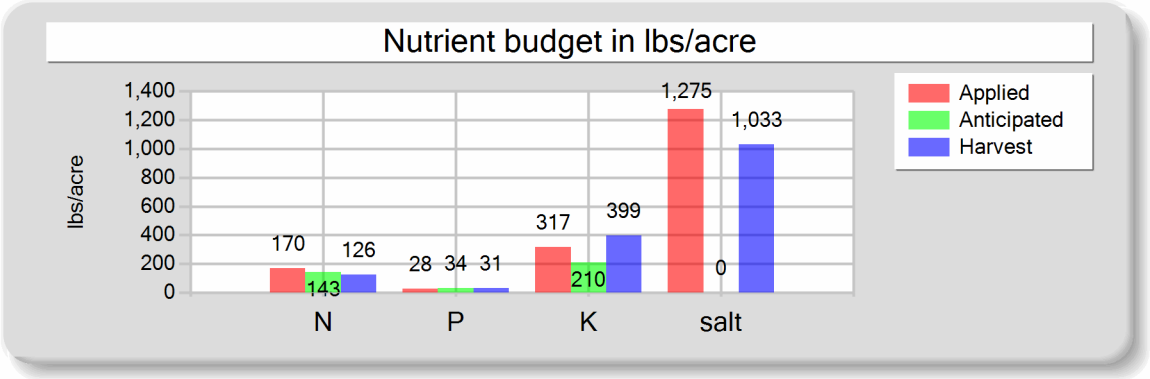
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
180,000.00 *gallons*
6.63 *acre-inches*
0.44 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #5 - 10/29/2022: Winter Forage

Field name: Field #5 Crop: Winter Forage Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	113.27	20.18	196.17	1,106.70
Process wastewater	49.57	8.01	121.00	168.57
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.84	28.19	317.17	1,275.27
Anticipated crop nutrient removal	142.80	33.60	210.00	0.00
Actual crop nutrient removal	126.26	30.65	398.54	1,032.90
Nutrient balance	43.58	-2.45	-81.37	242.36
Applied to removed ratio	1.35	0.92	0.80	1.23

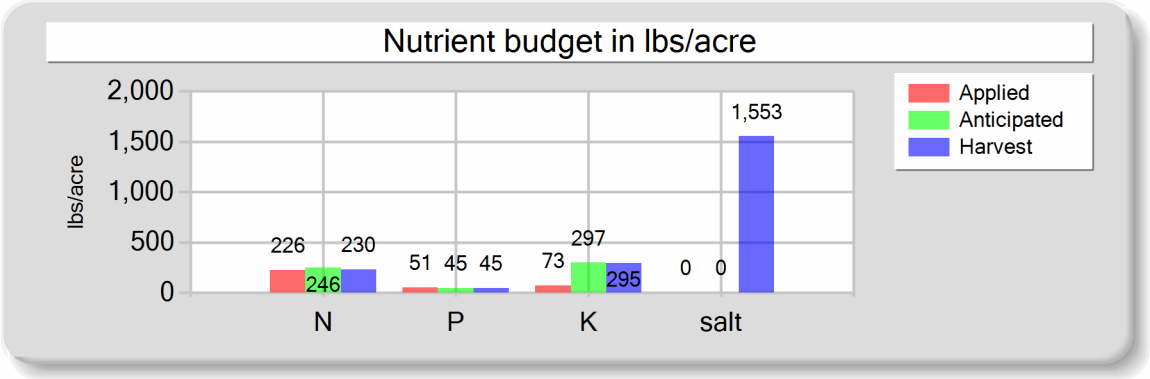
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
900,000.00 *gallons*
33.14 *acre-inches*
0.37 *inches/acre*

Total harvests for the crop
3 *harvests*

Field #5 - 06/20/2023: Corn, silage

Field name: Field #5 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	205.38	51.31	72.64	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.34	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.72	51.31	72.64	0.00
Anticipated crop nutrient removal	246.40	44.80	296.80	0.00
Actual crop nutrient removal	230.39	45.42	294.67	1,552.58
Nutrient balance	-4.66	5.89	-222.02	-1,552.58
Applied to removed ratio	0.98	1.13	0.25	0.00

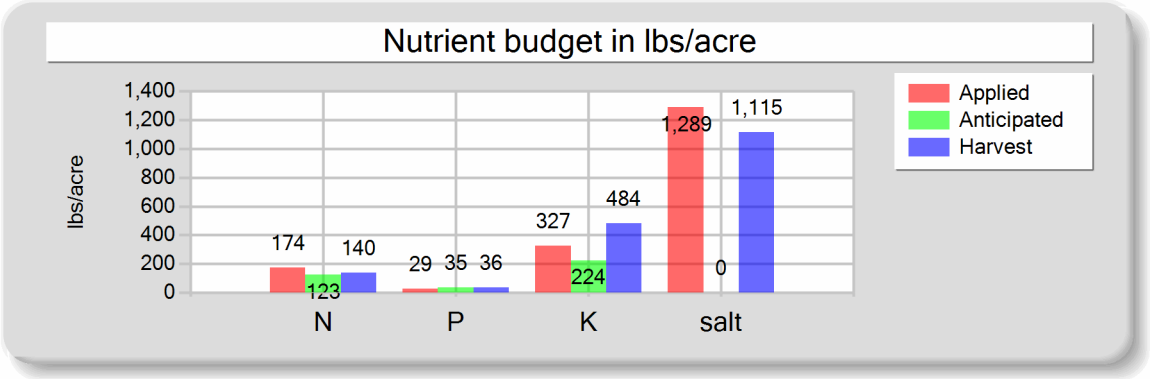
Fresh water applied	
65,058,000.00	*gallons*
2,395.86	*acre-inches*
26.62	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
3	*harvests*

Field #6 - 10/29/2022: Winter Forage

Field name: Field #6 Crop: Winter Forage Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	113.27	20.18	196.17	1,106.70
Process wastewater	53.53	8.65	130.68	182.05
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.81	28.83	326.85	1,288.75
Anticipated crop nutrient removal	123.20	35.00	224.00	0.00
Actual crop nutrient removal	139.51	36.38	484.02	1,114.53
Nutrient balance	34.30	-7.55	-157.17	174.23
Applied to removed ratio	1.25	0.79	0.68	1.16

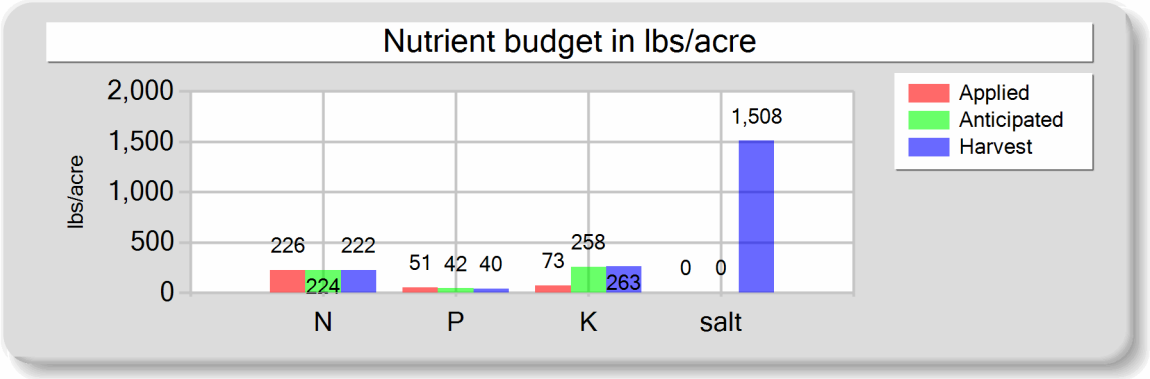
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
0.40	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #6 - 06/20/2023: Corn, silage

Field name: Field #6 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	205.38	51.31	72.64	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.56	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.95	51.31	72.64	0.00
Anticipated crop nutrient removal	224.00	42.00	257.60	0.00
Actual crop nutrient removal	222.25	40.35	262.87	1,507.90
Nutrient balance	3.70	10.96	-190.23	-1,507.90
Applied to removed ratio	1.02	1.27	0.28	0.00

Fresh water applied	
36,750,000.00	*gallons*
1,353.38	*acre-inches*
27.07	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2350015B

Sample and source description: T2350015B

Sample date: 12/22/2022 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,100.00	4,650.00	45,200.00	12,200.00	1,770.00	3,020.00	3,400.00	867.00		25.50
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3132032A

Sample and source description: T3132032A

Sample date: 05/12/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,100.00	7,770.00	11,000.00							0.00
DL	1.00	1.00	1.00							1.00

T3314022B

Sample and source description: T3314022B

Sample date: 11/10/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,110.00	3,540.00	10,600.00							0.00
DL	1.00	1.00	1.00							1.00

B. PROCESS WASTEWATER ANALYSES

T3053016A

Sample and source description: T3053016A

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	594.00	358.00	0.00	0.00	96.00	1,450.00								8,440.00	2,020
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3132032B

Sample and source description: T3132032B

Sample date: 05/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,300.00	549.00	0.00	0.00	122.00	1,620.00								9,430.00	9,400
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3314022A

Sample and source description: T3314022A

Sample date: 11/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	585.00	381.00	0.00	0.00	37.70	994.00								6,020.00	4,820
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

Domestic Dairy #1

Domestic Dairy #1

T3335009A

Sample description: T3335009A

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.14	0.00	1.14								248.00	0
DL	1.00	0.50	1.00								100.00	1

Domestic Dairy #2

T3335009B

Sample description: T3335009B

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.87	0.00	1.87								259.00	0
DL	1.00	0.50	1.00								100.00	1

Irr. Ag Well #1

T3335009C

Sample description: T3335009C

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.88	0.00	1.88								526.00	0
DL	1.00	0.50	1.00								100.00	1

Irr. Ag Well #2

Irr. Ag Well #2

T3335009D

Sample description: T3335009D

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.46	0.00	2.46								329.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field #1 - 10/29/2022: Oats, silage-soft dough

2395872

Sample and source description: 2395872

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,784.00	3,300.00	25,100.00		9.94
DL	1.00	1.00	1.00		1.00

Field #2 - 10/29/2022: Oats, silage-soft dough

Field #2 - 10/29/2022: Oats, silage-soft dough

2395873

Sample and source description: 2395873

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,904.00	2,700.00	30,500.00		10.31
DL	1.00	1.00	1.00		1.00

Field #3 - 10/29/2022: Oats, silage-soft dough

2395873

Sample and source description: 2395873

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,904.00	2,700.00	30,500.00		10.31
DL	1.00	1.00	1.00		1.00

Field #4 - 10/29/2022: Oats, silage-soft dough

2395872

Sample and source description: 2395872

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,784.00	3,300.00	25,100.00		9.94
DL	1.00	1.00	1.00		1.00

Field #5 - 10/29/2022: Winter Forage

T3132032C

Sample and source description: T3132032C

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	3,900.00	49,900.00		13.20
DL	1.00	1.00	1.00		1.00

T3132032D

Sample and source description: T3132032D

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	4,000.00	52,200.00		13.50
DL	1.00	1.00	1.00		1.00

T3132032E

Sample and source description: T3132032E

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,700.00	4,500.00	61,700.00		14.80
DL	1.00	1.00	1.00		1.00

Field #5 - 06/20/2023: Corn, silage

Field #5 - 06/20/2023: Corn, silage

T3293029A

Sample and source description: T3293029A

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,500.00	16,800.00		8.20
DL	1.00	1.00	1.00		1.00

T3293029B

Sample and source description: T3293029B

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,380.00	15,000.00		8.30
DL	1.00	1.00	1.00		1.00

T3293029C

Sample and source description: T3293029C

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	2,480.00	15,600.00		9.20
DL	1.00	1.00	1.00		1.00

Field #6 - 10/29/2022: Winter Forage

Field #6 - 10/29/2022: Winter Forage

T3132032E

Sample and source description: T3132032E

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,700.00	4,500.00	61,700.00		14.80
DL	1.00	1.00	1.00		1.00

T3132032F

Sample and source description: T3132032F

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,100.00	5,100.00	66,000.00		14.60
DL	1.00	1.00	1.00		1.00

Field #6 - 06/20/2023: Corn, silage

T3293029C

Sample and source description: T3293029C

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	2,480.00	15,600.00		9.20
DL	1.00	1.00	1.00		1.00

Field #6 - 06/20/2023: Corn, silage

T3293029D

Sample and source description: T3293029D

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,190.00	14,400.00		8.20
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

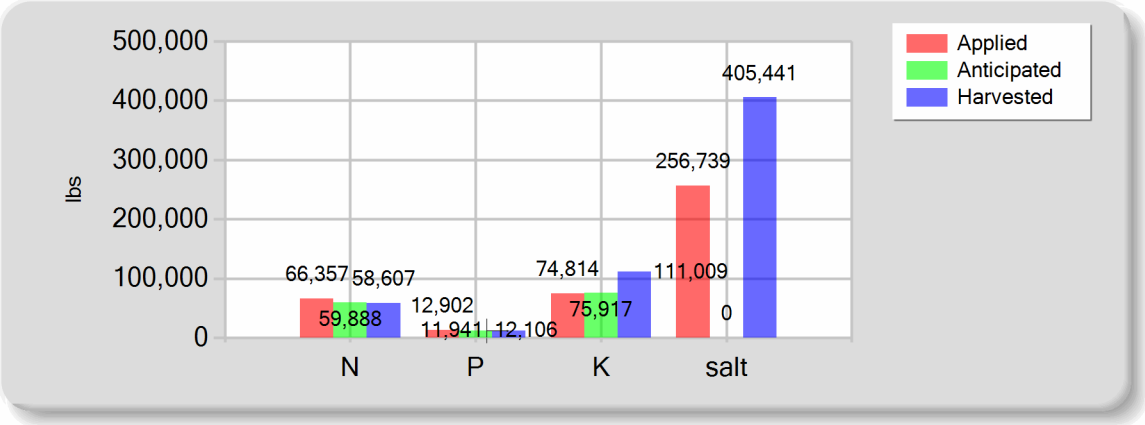
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

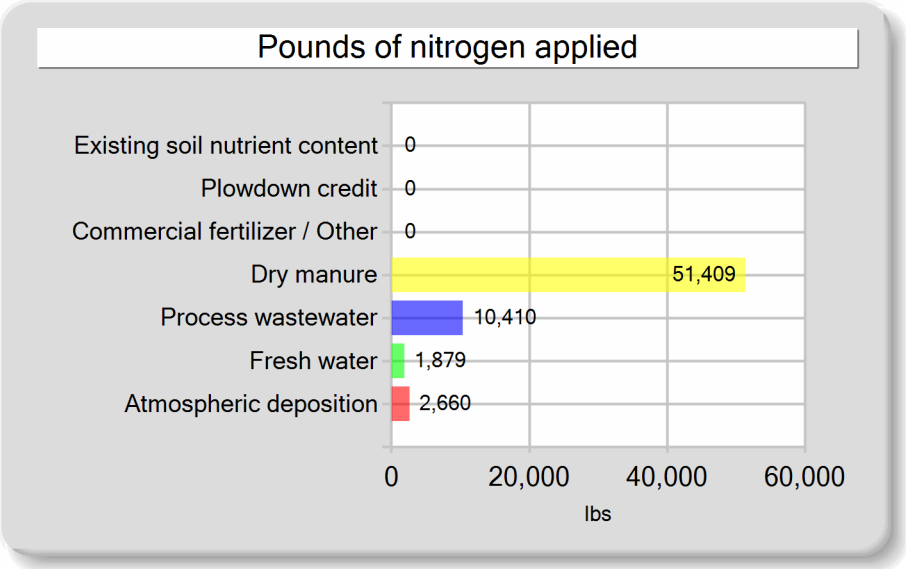
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

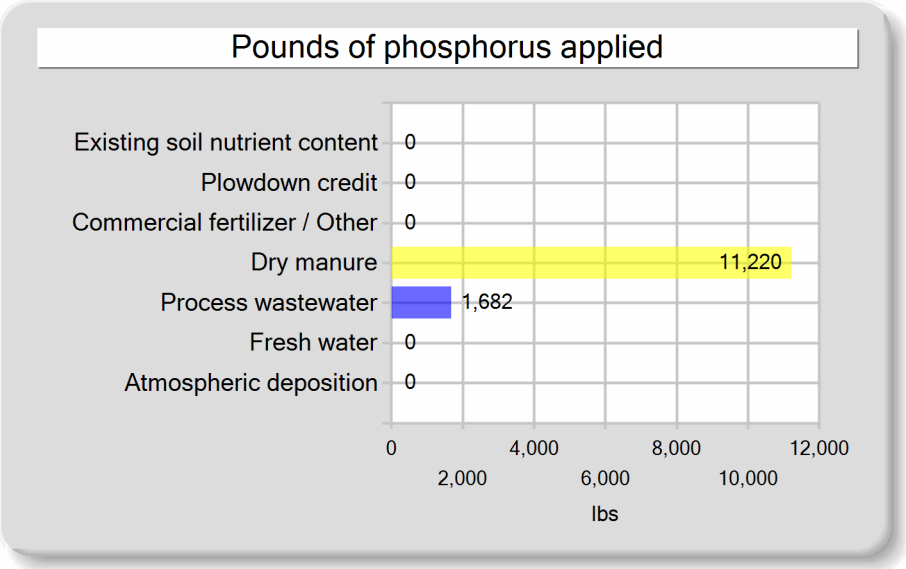
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	51,408.62	11,220.03	49,403.76	221,340.00
Process wastewater	10,409.55	1,682.35	25,410.53	35,399.49
Fresh water	1,878.80	0.00	0.00	0.00
Atmospheric deposition	2,660.00	0.00	0.00	0.00
Total nutrients applied	66,356.97	12,902.38	74,814.29	256,739.49
Anticipated crop nutrient removal	59,888.00	12,106.00	75,917.00	0.00
Actual crop nutrient removal	58,606.76	11,940.90	111,009.14	405,441.07
Nutrient balance	7,750.21	961.48	-36,194.86	-148,701.58
Applied to removed ratio	1.13	1.08	0.67	0.63

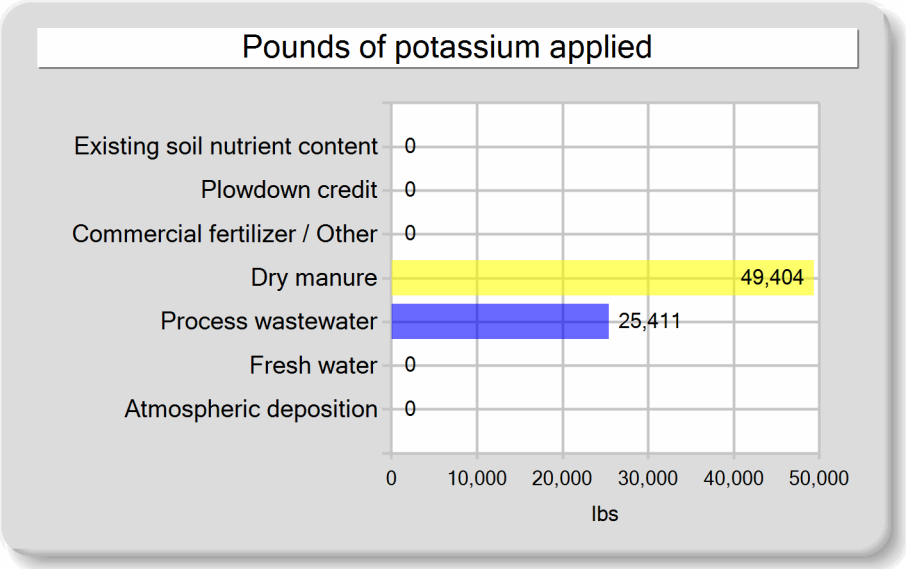
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

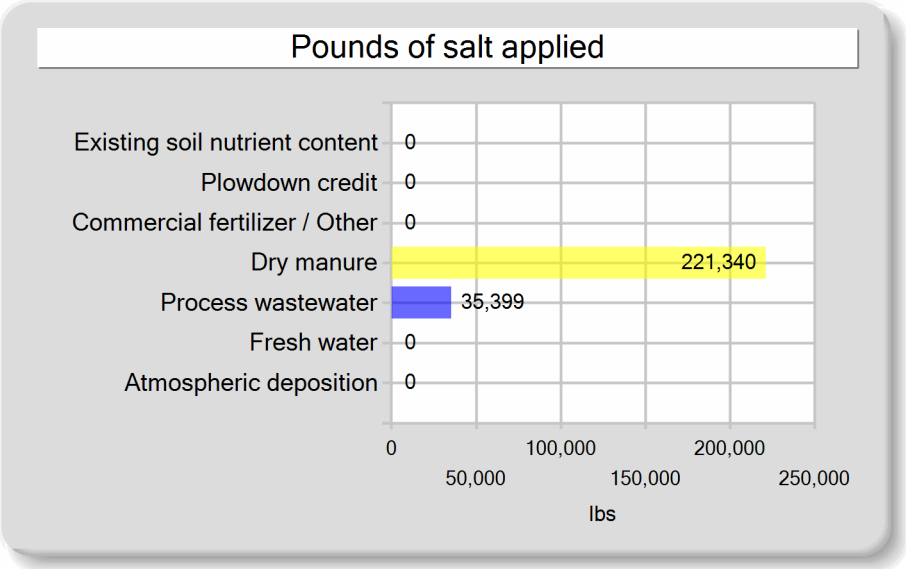


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

William J Van Ryn	Anthony B. Brazil
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
William John Van Ryn	Anthony Bartholomew Brazil
PRINT OR TYPE NAME	PRINT OR TYPE NAME
1-17-2024	1-22-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: BRAZIL/VAN RYN DAIRY FARM
Mailing address: P.O. BOX 149
(not facility address) RIPON, CA 95366
Phone#: (209) 482-2295

Authorized Copy Released To: JCRK, INC.
(209) 327-0992
Sampled By: Bill Van Ryn
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	11/30/23	8:00a	Domestic Dairy #1	Dom. Well	15.2	20 min	X				
2	✓	8:25a	Domestic Dairy #2	✓	15.6	✓	X				
3	✓	9:00a	Irr. Ag Well #1	Ag Well	15.4	✓	X				
4	✓	9:30a	Irr. Ag Well #2	✓	15.2	✓	X				
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	Jacquie Captain	JCRK, INC.		12/1/23 8:00a
2nd	Lana Lopez	DENELE ANALYTICAL, INC.	12-01-23 14:40	

(LABORATORY USE ONLY)

Logged In By:	LABORATORY NUMBER
LG	T3335009

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Brazil/Van Ryn Dairy Farm
PO Box 149
Ripon, CA 95366

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/1/23 2:40 PM
SUBMITTED BY: DENELE COURIER
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Dairy #1	T3335009A	Electrical Conductivity (EC)	EPA 120.1 and	0.248	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.14	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	158	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.8			
Electrical Conductivity (EC)	95.8			0.040
Nitrate Nitrogen (NO3-N)	95.3	109 (T3335007C)	105 (T3335007C)	3.39

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Customer
Brazil/Van Ryn Dairy Farm
PO Box 149
Ripon, CA 95366

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/1/23 2:40 PM
SUBMITTED BY: DENELE COURIER
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Dairy #2	T3335009B	Electrical Conductivity (EC)	EPA 120.1 and	0.259	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.87	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	165	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.8			
Electrical Conductivity (EC)	95.8			0.040
Nitrate Nitrogen (NO3-N)	95.3	109 (T3335007C)	105 (T3335007C)	3.39

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Compliance Analysis Report

Customer
Brazil/Van Ryn Dairy Farm
PO Box 149
Ripon, CA 95366

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/1/23 2:40 PM
SUBMITTED BY: DENELE COURIER
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irr. Ag Well #1	T3335009C	Electrical Conductivity (EC)	EPA 120.1 and	0.256	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.88	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	164	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.8			
Electrical Conductivity (EC)	95.8			0.040
Nitrate Nitrogen (NO3-N)	95.3	109 (T3335007C)	105 (T3335007C)	3.39

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Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/1/23 2:40 PM
SUBMITTED BY: DENELE COURIER
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irr. Ag Well #2	T3335009D	Electrical Conductivity (EC)	EPA 120.1 and	0.329	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.46	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	211	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.8			
Electrical Conductivity (EC)	95.8			0.040
Nitrate Nitrogen (NO3-N)	95.3	109 (T3335007C)	105 (T3335007C)	3.39

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

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